Controlling $2+2\to4$ Circuit Formation with a Parent-Only First Block

Summary. We derive a zero-temperature stationary phase diagram for a two-block residual ReLU network with independent weight decays, under the explicit restriction that block 1 produces the two supervised $d2$ parents but no $d4$ feature. At fixed representation strengths (s, q), we analytically eliminate the parent readouts, the block-2 target-producing weights, and the $d4$ readout. We then minimize the resulting scalar free energy over (s, q) separately for the direct $1+1+1+1 \to 4$ and compositional $2+2 \to 4$ circuits. The familiar condition $Q_2/Q_1^{(1)} > 1/\sqrt{8}$ selects composition only at a fixed representation; the phase of the trainable model is instead determined by comparing the two complete minima of $\Phi + G$.

1. Model and assumptions

For target parity $\chi_a(x) \in \{-1, +1\}$, let $f_a(x) = w_a^\top h_2(x)$ and define its target-mode amplitude $\widehat{A}_a = \mathbb{E}_x[f_a(x)\chi_a(x)]$. Under the pure aligned-mode ansatz $f_a = \widehat{A}_a\chi_a$, the sample loss reduces as $\mathbb{E}_x[(f_a - \chi_a)^2] = (1 - \widehat{A}_a)^2$, since $\chi_a^2 = 1$. The network and reduced weighted objective are therefore

$$\begin{aligned}
&h_1 = h_0 + \sigma(V_1 h_0), \qquad h_2 = h_1 + \sigma(V_2 h_1),\\
&f_a = w_a^\top h_2,\\
&\mathcal{J} = \frac{\gamma_2}{2}\sum_{a\in\{12,34\}}(1-\widehat{A}_a)^2 + \frac{\gamma_4}{2}(1-\widehat{A}_4)^2 + \frac{1}{2}\big(\lambda_E\|E\|^2 + \lambda_1\|V_1\|^2 + \lambda_2\|V_2\|^2\\
&\qquad\qquad + \lambda_p\|W_{d2}\|^2 + \lambda_4\|w_4\|^2\big)\,.
\end{aligned} \tag{1}$$

The derivation assumes: (A1) the full four-bit cube and pure ReLU; (A2) four equal-norm, orthogonal singleton Walsh vectors of norm $s = Q_1^{(0)}$; (A3) two equal-norm, orthogonal parent vectors q_{12}, q_{34} of norm $q = Q_2$; (A4) balanced rank-one gates, sufficient width, and additive parameter norms for parent and target sectors; (A5) the block-2 target vector aligns with its $d4$ readout; (A6) block 1 is restricted to the parent sector and produces no $d4$ feature; and (A7) the global zero-temperature minimum is reached. Thus this is an equilibrium mode theory, not a claim about finite-time SGD or dynamical lock-in.

The symbol V_1 in Eq. (1) is the *entire* first-block weight matrix. Under assumptions A4 and A6, we decompose it conceptually as

$$V_1 = V_{1,\text{parent}} + V_{1,\text{other}}, \qquad \|V_1\|_F^2 = \|V_{1,\text{parent}}\|_F^2 + \|V_{1,\text{other}}\|_F^2. \tag{2}$$

$V_{1,\text{parent}}$ is not a separate network parameter. It denotes the minimum-norm component of V_1 that implements the two supervised $1+1 \to 2$ parent gates. Assumption A6 removes the block-1 $d4$ sector rather than assigning it a finite cost; this is the simplifying restriction that avoids lower-mode leakage from a block-1 four-input ReLU.

The singleton byproduct of an ideal parent gate is not a fitted parameter:

$$\sigma[a(x_1 + x_2)] = \frac{a}{2}(1 + x_1 + x_2 + x_1x_2). \tag{3}$$

It has the same singleton and pair coefficient. If this generated singleton vector is orthogonal to the residual singleton vector, then

$$Q_1^{(1)} = \sqrt{s^2 + q^2}. \tag{4}$$

For overlap ρ, replace $s^2 + q^2$ by $s^2 + q^2 + 2\rho sq$. All gate-selection statements below remain exact when expressed using the measured $Q_1^{(1)}$.

2. Derivation roadmap: what is minimized, and in what order

The calculation is an algebraic elimination of weights from one simultaneous objective, not a claim that training executes the following steps in time. For each candidate block-2 target circuit $r \in \{D, C\}$, the minimization is organized as follows.

1. Choose nonnegative order parameters (s, q), where s fixes the four singleton-feature norms and q fixes the two $d2$ parent-feature norms.

2. Holding (s, q) fixed, minimize the $d2$ losses and the penalties on E, $V_{1,\text{parent}}$, and W_{d2}. This gives the parent-sector free energy $\Phi(s, q)$.

3. Still holding (s, q) fixed, choose either the direct block-2 circuit D or the compositional block-2 circuit C. Minimize its hidden $d4$ norm and readout norm. This gives the target-sector free energy $G(\kappa_r)$.

4. Finally minimize $\Phi(s, q) + G(\kappa_r)$ over (s, q). The two families perform this outer minimization separately, so their optimal representations need not be the same. The stationary phase is the family with lower final minimum.

Schematically, the nested calculation is

$$F_r = \min_{s,q\geq 0} \left[\underbrace{\min_{B,\, E, V_{1,\text{parent}}, W_{d2}} \mathcal{J}_{d2}}_{\Phi(s,q)} + \underbrace{\min_{Q\geq 0,\, w_4} \mathcal{J}_{d4}}_{G(\kappa_r)} \right], \tag{5}$$

where B is the common fitted amplitude of the two parent targets and $Q = Q_{d4}^{(2)}$ is the norm of the $d4$ coefficient vector produced by block 2.

3. Eliminating weights: the two stationary free energies

Equation (1) specifies the penalties but does not by itself imply the norms below; they are the minimum-norm realizations under A1–A4. Write e_i for the four orthogonal singleton vectors. Then $\|E\|_F^2 = \sum_i \|e_i\|^2 = 4s^2$. For one balanced $1+1 \to 2$ gate of hidden norm u, the preactivation amplitude per parent is $us/\sqrt{2}$; Eq. (3) therefore gives $q = us/(2\sqrt{2})$, or $u^2 = 8q^2/s^2$. The two disjoint parent gates add in squared norm. Finally, the minimum-norm readout satisfying $w_{ij}^\top q_{ij} = B$ is $w_{ij} = Bq_{ij}/q^2$, with $\|w_{ij}\|^2 = B^2/q^2$. Thus the two readout rows also add, giving

$$\|E\|^2 = 4s^2, \qquad \|V_{1,\text{parent}}\|^2 = \frac{16q^2}{s^2}, \qquad \|W_{d2}\|^2 = \frac{2B^2}{q^2}. \tag{6}$$

At fixed (s, q), the parent-sector objective is therefore

$$\Phi(s, q) \equiv \min_{B\geq 0} \left\{ \gamma_2(1 - B)^2 + 2\lambda_E s^2 + 8\lambda_1 \frac{q^2}{s^2} + \lambda_p \frac{B^2}{q^2} \right\}. \tag{7}$$

This definition includes both $d2$ losses and the regularization needed to create and read out the singleton and parent representations. It excludes every $d4$-specific weight and the $d4$ loss. Minimizing the displayed expression over the common parent amplitude gives

$$\begin{aligned} B^*(q) &= \frac{\gamma_2 q^2}{\gamma_2 q^2 + \lambda_p}, \\ \Phi(s, q) &= 2\lambda_E s^2 + 8\lambda_1 \frac{q^2}{s^2} + \frac{\gamma_2 \lambda_p}{\gamma_2 q^2 + \lambda_p}. \end{aligned} \tag{8}$$

The two allowed target circuits are now both in block 2: D is the direct $1+1+1+1 \to 4$ circuit and C is the compositional $2 + 2 \to 4$ circuit. Their efficiencies per target-producing hidden-weight norm u_r are

$$Q = \eta_r u_r, \qquad \eta_D = \frac{Q_1^{(1)}}{8} = \frac{\sqrt{s^2 + q^2}}{8}, \qquad \eta_C = \frac{Q_2^{(1)}}{2\sqrt{2}} = \frac{q}{2\sqrt{2}}. \tag{9}$$

Because the hidden penalty is $\lambda_2 u_r^2/2$, producing a hidden $d4$ norm Q costs $\kappa_r Q^2/2$, where the cost per squared target-mode norm is

$$\kappa_r \equiv \frac{\lambda_2}{\eta_r^2}, \qquad \kappa_D = \frac{64\lambda_2}{s^2 + q^2}, \qquad \kappa_C = \frac{8\lambda_2}{q^2}. \tag{10}$$

Thus smaller κ_r means a more efficient circuit.

Let $t = \|w_4\|$. Under the aligned pure-mode ansatz, the fitted target amplitude is tQ, so at fixed (s, q) the target objective is

$$\mathcal{J}_{4,r}(Q, t) = \frac{\gamma_4}{2}(1 - tQ)^2 + \frac{\lambda_4}{2} t^2 + \frac{\kappa_r}{2} Q^2. \tag{11}$$

Minimizing first over the readout norm gives

$$t^*(Q) = \frac{\gamma_4 Q}{\gamma_4 Q^2 + \lambda_4}, \qquad \min_t \mathcal{J}_{4,r}(Q,t) = \frac{\kappa_r}{2} Q^2 + \frac{\gamma_4 \lambda_4}{2(\gamma_4 Q^2 + \lambda_4)}. \tag{12}$$

The remaining minimization over $Q \geq 0$ yields

$$(Q_r^*)^2 = \left[\sqrt{\frac{\lambda_4}{\kappa_r}} - \frac{\lambda_4}{\gamma_4} \right]_+ . \tag{13}$$

The target is active precisely when

$$\kappa_r \lambda_4 < \gamma_4^2; \qquad \kappa_r \lambda_4 \geq \gamma_4^2 \implies Q_r^* = 0. \tag{14}$$

Substitution gives the target-sector free energy directly in terms of the circuit cost:

$$G(\kappa_r) = \begin{cases} \sqrt{\kappa_r \lambda_4} - \dfrac{\kappa_r \lambda_4}{2\gamma_4}, & \kappa_r \lambda_4 < \gamma_4^2, \\ \dfrac{\gamma_4}{2}, & \kappa_r \lambda_4 \geq \gamma_4^2. \end{cases} \tag{15}$$

The corresponding fitted target amplitude is

$$A_r^* = t_r^* Q_r^* = \left[1 - \frac{\sqrt{\kappa_r \lambda_4}}{\gamma_4} \right]_+ . \tag{16}$$

The closed phase theory is therefore

$$\boxed{F_D = \min_{s,q \geq 0} \{\Phi(s,q) + G(\kappa_D(s,q))\}, \qquad F_C = \min_{s,q \geq 0} \{\Phi(s,q) + G(\kappa_C(s,q))\}.} \tag{17}$$

Each phase chooses its own representation; the joint-training transition $F_C = F_D$ can therefore be first order even though the fixed-representation crossover is simple.

4. Phase diagram and control laws

At fixed (s,q), G is nondecreasing in κ, so the cheaper block-2 circuit is selected by

$$\kappa_C < \kappa_D \quad \Longleftrightarrow \quad \frac{q}{\sqrt{s^2+q^2}} = \frac{Q_2}{Q_1^{(1)}} > \frac{1}{\sqrt{8}}. \tag{18}$$

This is only a conditional, fixed-representation crossover. In the trainable model, the two circuits generally choose different minimizers (s_D, q_D) and (s_C, q_C). The actual phase boundary is therefore

$$\boxed{F_C = F_D,} \tag{19}$$

with F_D, F_C from Eq. (17). The ratio test in Eq. (18) becomes the full boundary only when (s,q) is frozen, or when Φ pins nearly the same representation for both circuits. For a learned compositional phase one must have

$$\boxed{F_C < F_D, \qquad \kappa_C(s_C, q_C)\lambda_4 < \gamma_4^2.} \tag{20}$$

The first condition compares the complete objectives after re-optimizing the representation; the second prevents decay from suppressing the $d4$ target. Equation (18) must also hold at a strict compositional optimum, but by itself it is not sufficient.

How to tune the restricted trainable model. (1) Decay w_4; without it, hidden/readout rescaling invalidates norm comparison. (2) Preserve the parents with strong $d2$ supervision and modest $\lambda_E, \lambda_1, \lambda_p$ until the *final* measured $Q_2/Q_1^{(1)}$ exceeds $1/\sqrt{8}$. (3) Avoid increasing λ_1 so far that the self-consistent parent norm q collapses; in this restricted model there is no block-1 $d4$ shortcut that benefits from large λ_1. (4) Tune $\lambda_2\lambda_4$ below the activity threshold while retaining nonzero decay on both factors. This controls target formation but does not alter the local $1/\sqrt{8}$ crossover. (5) Compare the complete measured objective of the direct and compositional solutions; the final accessibility ratio alone is not a global phase test when all parent weights remain trainable.

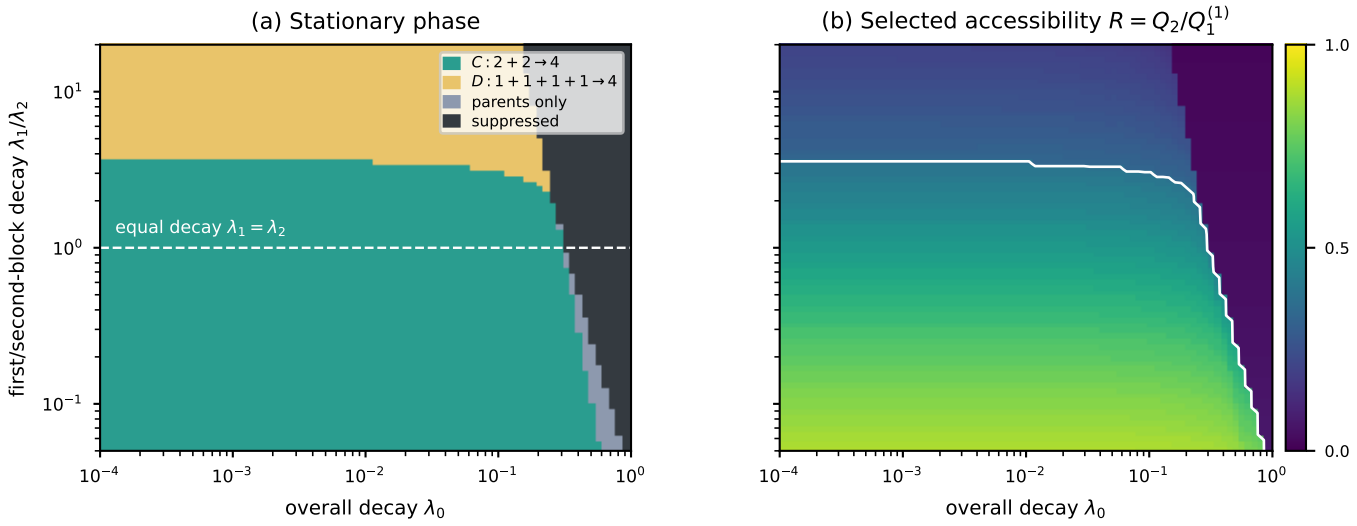


Figure 1: **Canonical stationary phase diagram.** We set $\gamma_2 = \gamma_4 = 2$, $\lambda_E = \lambda_p = \lambda_4 = \lambda_2 = \lambda_0$, λ_1/λ_2 as the vertical control, and use assumptions A1–A7. (a) Global minima of Eq. (17). Excessive first-block decay destroys the parents and favors the direct circuit; sufficiently large overall decay suppresses one or both tasks. (b) Accessibility $R = Q_2/Q_1^{(1)}$ at the selected global minimum. The white contour marks the fixed-representation crossover $R = 1/\sqrt{8}$; the global phase boundary need not coincide with it because the two families re-optimize (s, q) separately.

Control	Desired movement	Limiting failure
$\lambda_1 \downarrow$	preserve or enlarge $d2$ parents	large block-1 norms
$\lambda_2 \downarrow$	lower both block-2 circuit costs	large block-2 norms
$\lambda_p \downarrow$ or $\gamma_2 \uparrow$	preserve $d2$ parents	weak parent regularization
$\lambda_4 \downarrow$	keep the $d4$ target active	large readout norm
$\lambda_E \downarrow$	preserve singleton sources	embedding norms grow
$\gamma_4 \uparrow$	relax the activity threshold	does not alone select C

Scope. Nonzero singleton overlap ρ, unequal parent norms, imperfect target alignment, or multifunctional neurons enlarge the order-parameter set. Allowing block 1 to produce $d4$ invalidates assumption A6 and introduces an additional competing route and its lower-mode byproducts. Moreover, literal raw-ReLU parent and target gates emit constant and lower-degree modes; the pure-mode ansatz assumes that unwanted output modes are projected out or cancelled, and does not include the cost of that cancellation. Finite-temperature Langevin dynamics replaces $F_C = F_D$ by $F_C - F_D + \frac{T}{2}\log(\det' H_C/\det' H_D) - T\log(g_C/g_D) = 0$ and rounds the transition. Ordinary full-batch SGD is effectively zero-noise but need not reach the global saddle, so metastability and training history must be checked experimentally.